

RESEARCH NOTE / TECHNICAL REPORT

Deep Blue Research Report

A restrained XeLaTeX template for technical and
academic writing

Your Name

Laboratory / Department / Institution

August 2026 / Template v1.1

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CHAPTER 1

Overview

This template keeps the cover visually assertive while the body remains quiet, spacious, and suitable for long-form technical reports.

■ 1.1 Design intent

The visual system uses a dark navy foundation, a soft blue hierarchy, and a restrained yellow accent. The strong contrast is intentionally concentrated on the cover. Inside the report, color is limited to headings, fine rules, links, and lightly shaded table rows.

The page geometry is wider than many thesis defaults: the main text and tables use most of the A4 width while retaining enough margin for comfortable reading and printing. Body text uses a serif face, while headings and report metadata use a sans-serif face for a clean technical appearance.

Recommended use

Use this template for research notes, architecture reports, experiment summaries, literature reviews, project plans, and internal technical documents. Keep body graphics restrained; let data and structure carry the emphasis.

■ 1.2 At-a-glance layout

18 mm

Left and right margins

1.32x

Default table row height

A4

Optimized page size

■ 1.3 A wide table example

The table below demonstrates the default row treatment: a pale-blue row alternates with an unfilled row. The fill is intentionally subtle so dense reports remain easy to scan without looking like dashboards.

Table 1.1: Example of a full-width research table.

Component	Purpose	Suggested content	Visual weight
Cover	Establish report identity	Title, subtitle, author, affiliation, version, and date	Strong
Chapter opener	Mark major structure	A short chapter label, a large title, and a fine divider	Moderate
Section heading	Support scanning	Number, short title, and a small yellow square accent	Light
Body text	Carry the argument	Compact paragraphs with generous line length and modest spacing	Neutral
Data table	Compare evidence	Wide columns, alternating pale-blue rows, concise headers	Light
Callout	Surface one key point	One restrained blue-tinted box, used sparingly	Light

CHAPTER 2

Components

The body components are deliberately conservative: strong enough to make hierarchy obvious, but quiet enough for long technical documents.

■ 2.1 Headings and hierarchy

Chapter titles use navy with a thin pale-blue rule. Section headings use a small yellow square only as a locator; there are no large colored banners in the body. Subsection headings step down to a smaller blue sans-serif style.

2.1.1 A representative subsection

This is the default body level for most analysis. Equations remain conventional and inherit the surrounding typography:

$$S = \frac{T_{\text{baseline}}}{T_{\text{proposed}}}, \quad (2.1)$$

where S is speedup, and the two terms denote the baseline and proposed execution times.

2.1.2 Lists

Lists use a small blue bullet and compact spacing. They work best when the items are genuinely parallel:

- Keep section titles short and informational.
- Prefer full-width tables for methodology, prior work, and experiment matrices.
- Reserve yellow for small accents rather than body backgrounds.

■ 2.2 Figures

Use normal figure placement and captions. The frame below is only a placeholder; replace it with your own PDF, SVG-converted PDF, PNG, or JPEG asset.

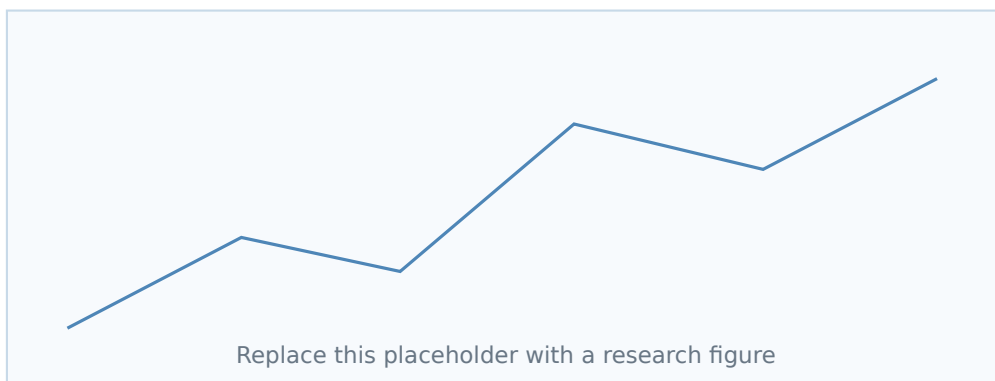


Figure 2.1: Restrained figure framing for the report body.

■ 2.3 Another table pattern

For dense comparative work, use the same alternating row pattern and let `tabularx` absorb the available width.

Table 2.1: Example methodology matrix.

Dimension	Setting	Rationale	What to report
Platform	Hardware / simulator	Establish reproducibility and architectural context	GPU model, clocks, memory system, interconnect
Workloads	Benchmark suite	Cover representative access patterns rather than a single kernel	Input size, warm-up, measured region
Baseline	Prior or default design	Avoid misleading speedups from weak comparison points	Exact configuration and source
Metrics	Performance and cost	Separate benefit from side effects	Speedup, bandwidth, coverage, accuracy, storage
Sensitivity	Key parameters	Test robustness instead of one tuned point	Sweep range and decision threshold

CHAPTER 3

Usage Guidance

Most users should only edit report metadata, section files, and the palette or geometry values in `report-theme.sty`.

■ 3.1 Project structure

The project is intentionally small and modular. The root file contains metadata and section order, while the visual system lives in one style file.

Table 3.1: Files included in the template.

Path	Role
<code>main.tex</code>	Entry point, report metadata, and section ordering.
<code>report-theme.sty</code>	Palette, typography, geometry, cover, headings, tables, headers, and helper components.
<code>sections/</code>	Example content split into chapter files. Replace these with your own report sections.
<code>figures/</code>	Recommended location for graphics and diagrams.
<code>Makefile</code>	Convenience commands for XeLaTeX compilation and cleanup.
<code>README.md</code>	Short setup and customization guide.

■ 3.2 Fast customization

For a slightly lighter report, change Navy and DeepBlue in the palette while leaving PaleBlue close to white. To make tables denser, reduce `\arraystretch` from 1.32 to around 1.18. To reclaim even more horizontal space, reduce the left and right margins from 18 mm to 16 mm; going much narrower is usually not recommended for printed A4 pages.

Body styling rule

Keep the cover as the main visual statement. Inside the report, prefer white space, typography, and alignment over additional colored panels. This preserves readability when the document grows from five pages to fifty.

■ 3.3 Compilation

Compile with XeLaTeX twice so the table of contents, page totals, and references settle correctly. The included Makefile performs this sequence.

CHAPTER 4

Code

Code blocks use the report's restrained colors while keeping language-aware syntax highlighting for source examples.

■ 4.1 Reusable code blocks

The codeblock environment accepts an optional title followed by an optional listings language. Use C, C++, or Python for syntax highlighting; leave the title empty when the code should appear without a heading.

C++ example

```
#include <numeric>
#include <vector>

int sum(const std::vector<int>& values) {
    return std::accumulate(values.begin(),
        values.end(), 0);
}
```

Python example

```
def normalize(values):
    total = sum(values)
    if total == 0:
        return values
    return [value / total for value in
        values]
```

The two examples use the same codeblock style; the titles identify the language while the two columns keep the examples together for comparison.

The report uses a BibTeX database with a numeric IEEE-style bibliography.

Bibtex Example[1]. The citation key is resolved from references.bib when the document is compiled with BibTeX.

Reference

- [1] J. E. Miller, H. Kasture, G. Kurian, C. Gruenwald III, N. Beckmann, C. Celio, J. Eastep, and A. Agarwal, "Graphite: A distributed parallel simulator for multicores," in *2010 IEEE 16th International Symposium on High-Performance Computer Architecture*. Bangalore, India: IEEE, Jan. 2010, pp. 1–12.